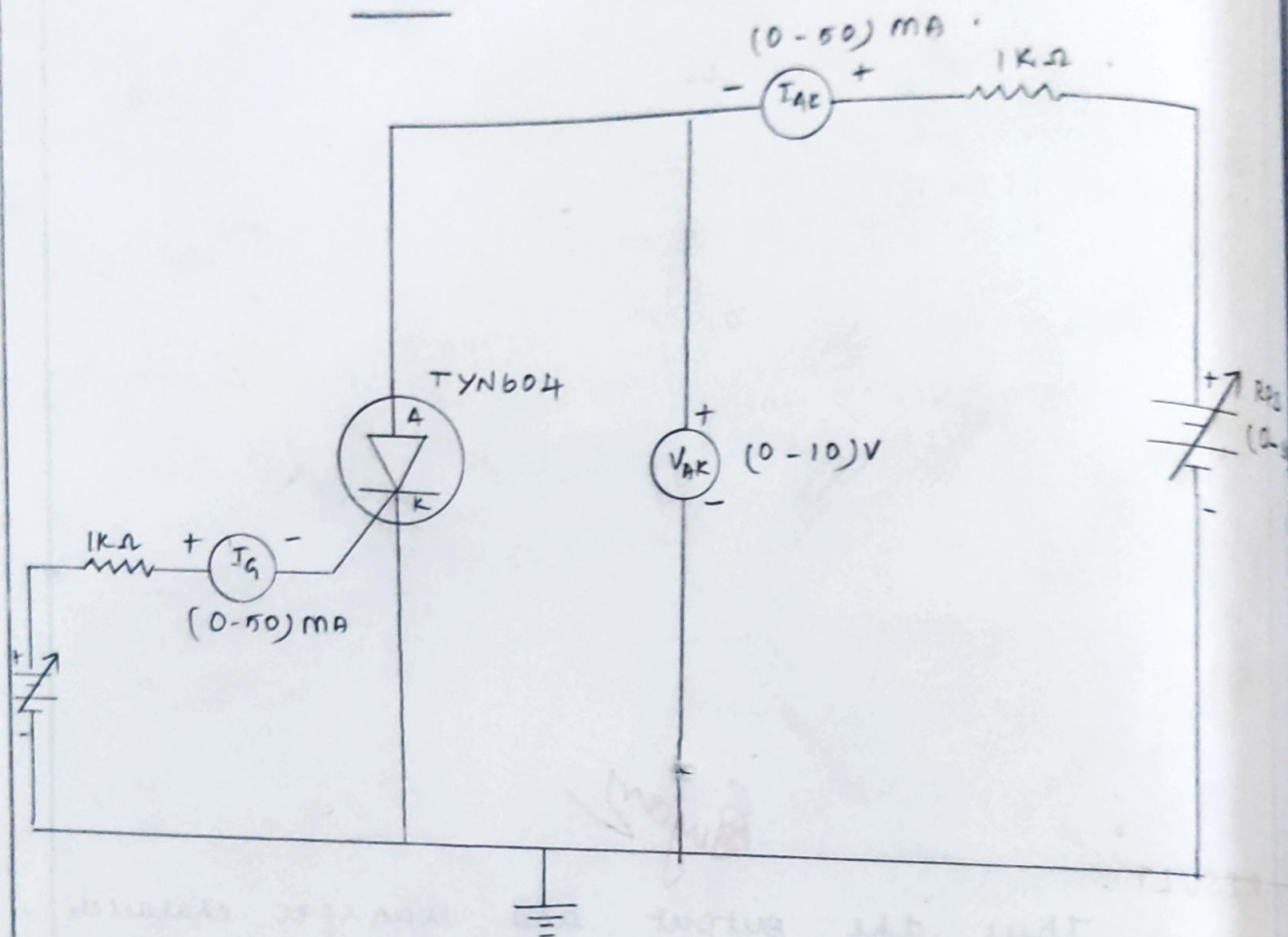


CIRCUIT DIAGRAM:

SCR



AIM:

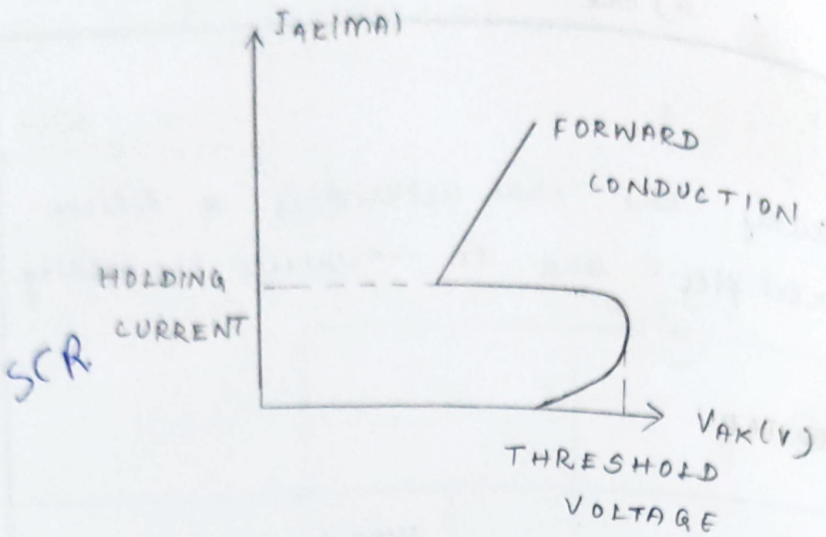
To study the characteristics of silicon controlled rectifier and to measure the holding current.

APPARATUS REQUIRED:

Sl. NO.	APPARATUS / COMPONENTS NAME	TYPE / RANGE	QUANTITY
1.	Dual Regulated Power supply (RPS)	(0-30) V	1
2.	Voltmeter	(0-300) V	1
3.	Ammeter	(0-20) mA	1
4.	Bread Board	-	1
5.	Silicon controlled Rectifier	2N3669	1
6.	Resistance (R_1)	470 Ω / 0.5 W	1
	(R_2)	100 Ω / 10 W	1
7.	connecting wire	-	As required

MODEL GRAPH:

SCR



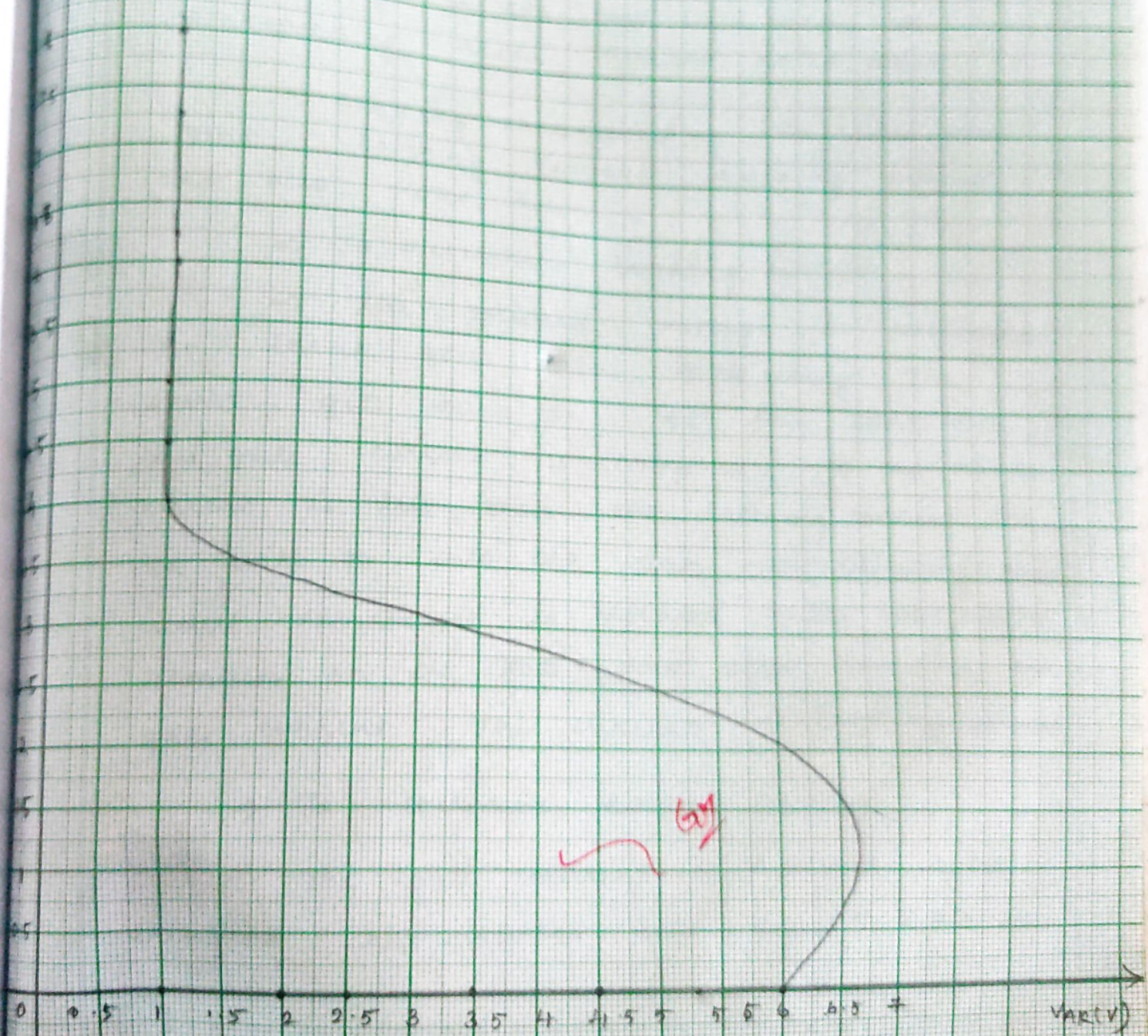
TABULATION :

$$I_G = 5.5 \text{ mA}$$

$V_{AK}(V)$	$I_{AK}(mA)$
1	0
2	0
2.5	0
3.4	0
4.5	0
5.3	0
6	0
1	4.5
1	5.9
1	6.3
1	7

FOR ...

FOR ...



PROCEDURE:

- ⇒ The connections are made as per the circuit diagram.
- ⇒ RPS is adjusted so the required gate current flows through the device.
- ⇒ The V_{AK} is increased in step by step by varying the RPS and each step, the corresponding anode current noted down.
- ⇒ Reading corresponding to break over point, latching current, holding current and also noted down.
- ⇒ To determine latching current.
- ⇒ The forward voltage is increased to break over value, gradually. The device gets turn ON. At this condition, the gate current is removed. If the anode current is less than the latching current the SCR will go into OFF state immediately. The circuit resistance is repeated until the device remain in ON state even after removing the gate current. This current is latching current.
- ⇒ To determine holding current.
- ⇒ The gate current is removed after bringing the device into the ON state. The



circuit resistance is increased steps. The minimum value of anode current at which the SCR remains the on state is the holding current.

RESULT :

Thus the characteristics of silicon controlled rectifier are determined.

(i.e) Holding current = 12.5mA

